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Modeling the Electrical Conductive Paths within All-Solid-State Battery Electrodes

All-solid-state batteries constitute a very promising energy storage device. Two very important properties of these battery cells are the ionic and the electrical conductivity, which describe the ion and the electron transport through the electrodes, respectively. In this work, a numerical method is presented to model the electrical conductivity, considering the outcome of discrete-element method simulations and the intrinsic conductivities of both the active material particles and the conductive additive particles. The results are calibrated and validated with the help of experimental data of real manufactured electrodes. The tortuosity, which strongly influences the ionic conductivity, is also presented for the analyzed electrodes, taking their microstructure into account.

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Keywords: All-solid-state battery electrodes, DEM simulations, Electrical conductivity, microstructure, Tortuosity

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1 Introduction

Even though conventional lithium-ion batteries (LIB) have gained widespread recognition especially in the field of consumer devices such as smartphones, laptops, and electric vehicles, there are still safety issues that must be confronted. Recent accidents involving fire or even explosions of LIB in smartphones and battery electric vehicles have been reported to be caused by the flammable liquid electrolyte [1–4]. Bearing this in mind, lithium-ion all-solid-state batteries (ASSB) are becoming more and more popular as they constitute a promising alternative to conventional LIB devices. In fact, ASSB are composed of solid electrolytes, increasing the safety of the battery by avoiding the risk of possible fluid leakage. Besides improving the battery safety, solid electrolytes offer the distinct possibility of new battery geometries, which may lead to a higher packing efficiency of the cells and, thus, an increased battery energy density. Another key aspect is the feasibility of directly using Li metal on the anode side since the risk of dendrite formation can be highly decreased if an appropriate solid electrolyte is used. As a result, the size of this novel battery system can be reduced substantially, increasing the energy density as well [5–7].

Despite their growing potential, there is still room for further enhancements, especially in terms of improving the energy and power densities. Taking this into consideration, conduction phenomena within the electrode, i.e., electrical and ionic conductivity, are two relevant features that must be well understood since they dominate the overall cell properties. In fact, it has already been reported for conventional LIB that electronic and ionic conductivities play significant roles during the intercalation and deintercalation of charge-carrying ions [8–10]. Nevertheless, little research has been conducted on these prop-

erties within the context of ASSB, so far. One of the few works has dealt with the ionic conductivity of the solid electrolyte, demonstrating the influence of various ASSB electrode structures on this parameter [11]. Hou et al. [12] confirmed the major significance of interconnected electronic pathways in the electrochemical performance of a lithium-sulfur ASSB and experimentally verified the equal relevance of electron conduction in comparison to ion transfer. Within this framework but adopting a numerical approach, this study proposes a novel method to determine the electrical conductivity of a given ASSB electrode structure previously generated by the discrete-element method (DEM). Additionally, this post-processing tool can be used to give insights into the fundamental role of the electrode structure. This is of considerable importance because a structure that enables a good electrical conductivity might at the same time hinder the ionic conductivity. Finding optimal electrode architectures that offset competing tradeoffs can be challenging and, therefore, there is a clear need to fully comprehend the influence of the electrode microstructure on the cell performance. Taking this into account, computational and simulation tools have been established as a non-invasive alternative and an extension to experiments. Some widely used models for simulating the electrical conductivity adopt a

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continuum-medium approach, which involves the homogenization of the electrode components, eliminating the need to describe the microstructure in detail [13–15]. As a matter of fact, Bruggeman's effective medium theory has been used to correlate and model effective transport, yielding quantitatively good predictions for conventional LIB [16, 17]. Another very well-known numerical discrete-medium approach is the random resistor network model approach, which is based on creating a resistance network and calculating the conductivity by imposing a potential difference along the specified direction and applying Kirchhoff's current law [18]. For instance, Schneider et al. [19] used this method to study the effects of some structural parameters on the electrical conductivity of monosized particulate solid oxide fuel cells (SOFC). Also in the field of SOFC, Abbaspour et al. [20] simulated electron and ion conductor particles to determine both ionic and electrical transfer processes. Both continuum-medium and discrete-medium theories have been subject to some improvements in connection with the percolation theory, which has been extensively used to model phase transition. A percolation model was developed by Chen et al. [21] to predict the effective properties of a composite electrode made of electronic and ionic conducting particles. Furthermore, they analyzed the effects of the particle size and the particle size distribution on the effective electrical conductivity and investigated the percolation volume fraction threshold as well. However, despite the extensive research dedicated to SOFC and the moderate research devoted to LIB, the application of numerical techniques to assess the electrical conductivity of ASSB electrodes is, to our knowledge, particularly scarce, so far. Bearing in mind the existing approaches, the goal of this study was to implement a method based on the random resistor model with the addition of an improvement in the computational cost due to the simplification of the microstructure. By generating ASSB electrodes via DEM simulations, this method enables assessing the impact of the electrode structure on the electrical paths. Moreover, the calibration and subsequent validation of the numerical results with real experiments enables analyzing the impact of the carbon black (CB) concentration on the specific electrical conductivity of the electrode. As it has been mentioned, a compromise between the electrical transport and the ionic transport must be sought during electrode production. For this reason, the last step was to calculate the tortuosity of the simulated structures in order to assess the ionic transport through the electrolyte and within the electrode.

2 Materials and Experimental Methods

In order to validate the numerical method, ASSB electrodes were manufactured with different amounts of CB. The electrodes consisted of lithium iron phosphate (LFP) as active material, CB (Imerys Graphite & Carbon Switzerland SA) as electrical conductive additive, and a solid electrolyte containing polyethylene oxide (PEO; POLYOXWSR 205, Dow) with a molecular weight of 600 g mol^{-1} and a concentration of 3M lithium *bis*(trifluoromethylsulfonyl)imide (LiTFSi) as conducting salt. The carbon content was varied between 0 and 5.2 wt %, but the proportion of active material to solid electro-

lyte was kept constant at 9.64:2.2:1 wt % (LFP/PEO/LiTFSi). The components were mixed through a melt granulation step in a Ploughshare[®] mixer L5 (Gebrüder Lödige Maschinenbau GmbH) for 30 min at 90 °C, with a slow runner rotational speed of 150 rpm and a chopper speed of 2500 rpm. The granules were further processed by extrusion with a twin screw extruder from Coperion GmbH at 120–150 °C and a mass flow of 0.5 kg h^{-1} and 150 rpm. Finally, a calendering step was carried out via a laboratory calender device (Saueressig GmbH) with a roll temperature of 90 °C and a roll speed of 0.1 m min^{-1} to reach the targeted electrode thickness of 60 µm. The electrode mass was determined with a laboratory scale and the thickness with an indicating gauge (Mitutoyo, Planolith, ID-C112AXB); both are required to calculate the porosity of the electrodes according to Froboese et al. [11]. The electrical conductivity measurements were performed via a two-point approach with a material testing machine (Z020; Zwick GmbH & Co. KG). Between two conducting stamps, the sample was analyzed by applying a certain load. For the determination of the intrinsic electrical resistivity of LFP and CB, a load of 380 N was applied to the powder sample to assure good electric contacts in the particle bed without damaging its structure. For the measurement of the specific electrical conductivity of the electrodes, the load was 40 N. In both cases, the electrical resistance was measured for 3 s, applying a current of 10 mA. The scanning electron microscopy (SEM) picture was taken with the SEM device Helios G4 CX (FEI Deutschland GmbH, Frankfurt am Main, Germany).

3 Computational Modeling

3.1 Discrete-Element Simulations of ASSB Electrodes

The ASSB electrode structures were generated using the open source DEM simulation software LIGGGHTS [22]. Even though this computational method was in fact not used for its real purpose, which is to analyze motion and effect of particles, it served to create the electrode structure. Besides, the DEM was chosen as the means to generate the particle network because it allows calculation of the contact area between overlapping particles. This is used by the subsequently implemented method to calculate the electrical resistance of the contacts, as it is shown in the following section. It is important to underline that the real measured porosity of the manufactured electrodes was in all cases negligible and practically zero, for all of them, mainly due to the last production step: the calendering process. As a result, the electrodes consisted of the electrolyte, active material particles, and conductive additive particles. Due to the fact that the aim of this study was to model the electrical conductivity and the employed PEO-LiTFSi electrolyte mixture is an electrical insulator, only LFP and CB particles were simulated. The electrolyte was therefore regarded as pores between the particles since it does not contribute to the determination of the electrical conductive paths within the structure. Thus, the porosity of the simulated electrodes corresponded in reality to the volume occupied by the electrolyte. For the representation of the particles and based on the laser diffraction measurements of the LFP powder, seven particle sizes and their

corresponding frequency distribution values were selected in the simulation (Tab. 1). Regarding the CB primary particles, they are much smaller than the LFP ones, as can be seen in Fig. 1. These SEM images clearly show the contrast between the two particle sizes. In fact, CB primary particles have a diameter of approximately 40–50 nm whereas LFP particles have a diameter of around 350 nm. However, the CB nanoparticles aggregate easily, forming strongly connected particle units. The mean diameter of a CB aggregate ranges normally between 85 and 500 nm [23]. In this work, a particle size of 65 nm was established, bearing in mind the strong shearing forces that appear within the mixing process for the production of the electrode paste and which promote aggregate fragmentation.

Table 1. Particle sizes of LFP and the cumulative distribution of their incremental increase as input parameters for the generation of the electrode structure in the DEM.

Particle size of the LFP [mm]	ΔQ_3 [%]
0.28	15.00
0.30	15.00
0.32	15.00
0.34	10.00
0.36	10.00
0.40	15.00
0.60	20.00

Due to computational limitations, it would be highly costly and non-efficient to simulate the whole electrode on the micro-scale. Consequently, a representative volume element (RVE)

was chosen to assure consistent results without compromising the simulation time. At the beginning of the simulations, particles were placed randomly in the RVE, assuring there was no overlap between them. No gravitational effects were contemplated and the boundaries were considered periodic in x - and y -direction. In order to reach the experimental volume content of LFP and CB and to ensure particle contacts, the assembly was compressed with a bottom and a top plate moving in opposite directions. The absolute velocity of each plate was set as equal to the velocity of the sample holder of the material testing machine used to measure the electrical conductivity of the electrodes, 0.75 mm s^{-1} , which is low enough to ensure no fundamental changes in the electrode structure. The volume content of the LFP and CB particles, $V_{\text{par}}^{1)}$, within the simulation domain was calculated along the process and considered as the criterion to stop the simulation once the experimental value was achieved. V_{par} was determined as

$$V_{\text{par}} = \frac{V_{\text{LFP}} + V_{\text{CB}}}{V_{\text{RVE}}} = \frac{V_{\text{LFP}} + V_{\text{CB}}}{lwh} \quad (1)$$

where V_{LFP} and V_{CB} are the volumes of the LFP and CB particles, respectively, and V_{RVE} is the volume of the representative volume element, calculated as the product of the length l , the width w and the height h of the simulated domain.

The contact model used for the characterization of the LFP and CB particles was the Hertz-Mindlin contact model, with the additional implementation of van der Waals forces to account for the short-range attractive interactions [24]. The van der Waals forces were calculated regarding the first derivative of the van der Waals potential [25]. Due to stability reasons, the action of these forces was limited to a distance value d , which depended on the particle size. Tab. 2 depicts the corresponding input parameters required for LFP and CB.

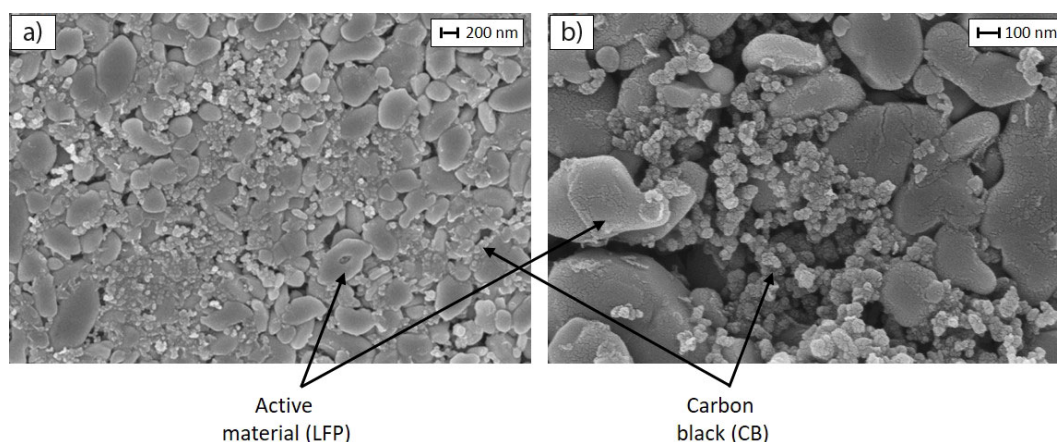


Figure 1. (a) Low- and (b) high-magnification SEM pictures of a CB-LFP granule: Smaller particles correspond to CB whereas the coarse ones correspond to the active material, LFP.

1) List of symbols at the end of the paper.

Table 2. Input parameters for characterization of the LFP and CB particles.

Parameter	LFP	CB	References
Young's modulus, E [GPa]	123.90	5.00	[26–28]
Poisson ratio, ν [–]	0.28	0.23	[26–28]
Coefficient of restitution, e [–]	0.40	0.40	[26,27]
Density, ρ [g cm ^{−3}]	4.28	1.96	from supplier
Distance limit, d [mm]	7.30×10^{-6}	7.80×10^{-6}	[25, 29–31]
Hamaker constant, A [10^{-19} J]	1.21	1.09	[25, 29–32]

3.2 Modeling the Specific Electrical Conductivity of ASSB Electrodes

The implemented tool to model the specific electrical conductivity was written in Python programming language and profits from the structural information delivered by the DEM simulation as explained in the following lines. Once the electrode structures were generated via DEM and the final LFP and CB content was achieved, a file was created containing the information related to every particle, including their particle identifier (ID) material, radius, and position within the electrode assembly. Another file regarding the particle contacts within the simulated electrode was also saved, containing the pair id of the particles in contact and the contact area involved, which was implemented explicitly for the purpose of this work. The first step of this post-processing method to assess the electrical conductivity consists in generating a graph with the information of these two files. This graph depicts the connectivity within the electrode structure and comprises nodes, which represent the particles, and connecting lines, which describe the Euclidian distance between two particles in contact.

In order to determine the electrically conductive paths within the simulated cathodes, a pathfinding algorithm was implemented that served to create the equivalent circuit in the following step. Within an ASSB, electrons have to move through the cathode particle network during loading and unloading of the battery. In fact, during the experimental measurement of the electrical conductivity of an electrode, an electric current is given and the created voltage difference is measured. Thus, for the computational study, we considered the lowest positioned particles as the start nodes and the particles with the highest z -position values as the goal nodes. As already mentioned, a pathfinding algorithm was implemented to imitate the movement of the electrons within the electrode. In general, a pathfinding algorithm searches a graph by starting at a certain position and examining adjacent nodes until the goal node is reached, usually with the purpose of detecting the shortest route. Taking the already developed algorithms into consideration [33,34], it was decided to adopt a heuristic approach, which is less time consuming due to the fact that it is not

exhaustive as it strategically eliminates impossible paths. In particular, the implemented algorithm in this study was based on the A* algorithm, well known in computer games and navigating systems. First published in 1968 [34], it is an extension of Dijkstra's algorithm and it mainly works by minimizing the function $f(n)$, with n being the next node on a path. $f(n)$ is calculated by summing two functions: a cost function that accounts for the cost of the path from the start node to n and a heuristic function that estimates the cost of the shortest path from n to the goal node, $g(n)$ and $h(n)$, respectively. The following equations show how to calculate the described functions:

$$f(n) = g(n) + h(n) \quad (2)$$

$$g(n) = g(n_{\text{current}}) + \|\vec{r}_{n_{\text{current}}} - \vec{r}_{n_{\text{next}2}}\| \quad (3)$$

$$h(n) = |z_{n_{\text{goal}}} - (z_{n_{\text{next}}} + r_{n_{\text{next}}})| \quad (4)$$

In order to reckon the cost function (Eq. 3), the Euclidian distance between the current node and the suggested next node is summed to the already calculated distance between the start and the current node, $g(n_{\text{current}})$. For this purpose, the algorithm works with two lists: an opened and a closed list, where all the analyzed nodes are classified to avoid examining a node twice. The heuristic function (Eq. 4) can never overestimate the distance to the goal node and is calculated considering the positions of the nodes in z -direction. The combination of both functions guarantees that the shortest path from all start nodes, i.e., particles, to the goal ones is accomplished. However, due to the difference in the intrinsic electrical resistivities of LFP and CB, the shortest path does not necessarily imply the most favorable path for the transfer of the electrons. Indeed, the intrinsic electrical resistivity of CB is almost 10^5 lower than that of LFP, as it is shown in the following section. Consequently, the A* algorithm had to be adjusted to make sure that the selected pathways corresponded to the electrically most favorable paths, taking into consideration the contact resistance between the particles. Eqs. 3 and 4 were modified accordingly:

$$g(n) = g(n_{\text{current}}) + \|\vec{r}_{n_{\text{current}}} - \vec{r}_{n_{\text{next}2}}\| \frac{\rho_{ij}}{A_{ij}} \quad (5)$$

$$h(n) = |z_{n_{\text{goal}}} - (z_{n_{\text{next}}} + r_{n_{\text{next}}})| \frac{\rho_i}{A_i} \quad (6)$$

where ρ_{ij} corresponds to the mean value of the electrical resistance of both particles engaged in the evaluated contact. This means that the electrical resistances of LFP and CB are also considered in the selection of the paths. A_{ij} corresponds to the contact area. Since the heuristic value has to be smaller than the most conductive path to the goal node, ρ_i corresponds to the resistivity of the most conductive material in the pairwise contact and A_i is the greatest contact area. With this correction, it can be guaranteed that the electron path is created that favors the connections through CB particles, which are more conductive than LFP particles. After running the algorithm, the position vectors of each particle, the contact areas, and the resistivity of each particle were saved for all resulting conductive paths.

The third step of the method towards determining the electrical conductivity consisted in devising an equivalent circuit with the information obtained by the adjusted A* algorithm. For this computation, the individual resistances were calculated by approximating the total resistance of a path with a series of fictional cylinders. The cylinder resistances were determined as shown in Fig. 2. Firstly, the coordinates of the contact points, $\vec{c}_{1,2}$, were implemented as:

$$\vec{c}_{1,2} = \vec{r}_1 + \frac{\vec{r}_2 - \vec{r}_1}{\|\vec{r}_2 - \vec{r}_1\|} \left(r_1 - \frac{\delta}{2} \right) \quad (7)$$

where the subscripts 1 and 2 corresponded to the particles involved in the contact. $\vec{r}_1$ and $\vec{r}_2$ are the corresponding position vectors of particles 1 and 2; r_1 is the particle radius and δ is the extent of the overlap. Subsequently, the Euclidean distance between two consecutive contact points, $\vec{c}_{1,2}$ and $\vec{c}_{1,3}$, could be calculated. This constituted the length of the cylinder, l_{cyl} :

$$l_{\text{cyl}} = \|\vec{c}_{1,2} - \vec{c}_{1,3}\| \quad (8)$$

The cross-sectional area of the cylinder was reckoned as the average area of the contacts:

$$\bar{A} = \frac{A_{1,2} + A_{1,3}}{2} \quad (9)$$

The resistance could be evaluated from the intrinsic resistivity of the particle and the mentioned length and transversal area of the determined cylinder:

$$R_p = \frac{\rho l_{\text{cyl}}}{\bar{A}} \quad (10)$$

The consideration of the contact area is extremely important since the quality of the interparticle contacts directly impacts on the electron conduction. The manufacturing steps and the operation of the battery affect this feature to a great extent. In

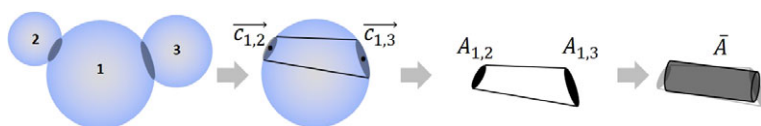


Figure 2. Approximation of the resistance by a cylinder.

fact, the greater the calendering level, the higher the area between the particles in contact. Moreover, volume changes of active material particles arisen during charge and discharge processes also alter the interparticle contacts, leading to a gradual loss of connectivity. These changes within the electrode structure are implicitly regarded in the presented model since an alteration in the quality of the interparticle contacts would result in different values of electrical conductivity.

The overall resistance of each selected conductive path could be computed out of the individual resistances connected in series. As a result, a total equivalent resistance could be calculated by connecting every path in parallel. Finally, the total specific electrical conductivity σ could be determined from the equivalent electrical resistance and the size of the simulated ASSB electrode as [35]:

$$\sigma = \frac{h_{\text{RVE}}}{w_{\text{RVE}} l_{\text{RVE}} R_{\text{eq}}} \quad (11)$$

Fig. 3 sums up this numerical procedure schematically. By all means, this method was followed taking all three space dimensions into account.

3.3 Tortuosity Determination and Ionic Conductivity Estimation of ASSB Electrodes

The tortuosity τ is commonly used along with the electrode porosity ε to quantify the diffusion transport process throughout an electrode. In geometrical terms, the tortuosity can be defined as the ratio between the shortest pathway connecting two points and their Euclidean distance [36]. Thus, bottlenecks of the pore structure are not considered following this definition. Vijayaraghavan et al. [37] proposed a relation between the global structural properties, i.e., the porosity, the tortuosity, the electrolyte diffusion coefficient D_{el} , and the total diffusion coefficient D_{eff} , which has been widely used for conventional LIB electrodes:

$$D_{\text{eff}} = \frac{\varepsilon}{\tau} D_{\text{el}} \quad (12)$$

For the tortuosity determination of the simulated electrode structures, the positions and sizes of all

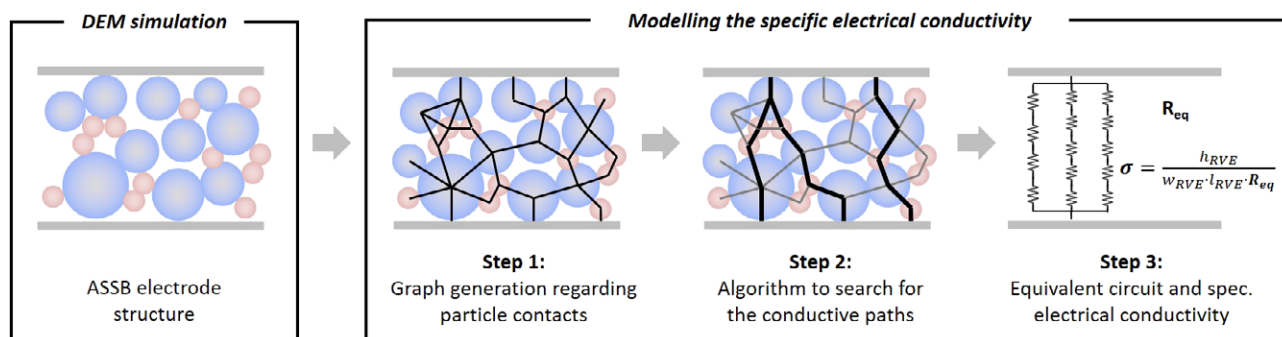


Figure 3. Two-dimensional schematic of the numerical procedure for the determination of the equivalent electrical conductivity of a simulated DEM electrode structure.

particles were used to define the pores via an implemented MATLAB code. Subsequently, a skeleton model was derived according to Lee et al. [38] and the tortuosity was calculated from the effective length of the path and the height of the simulated domain as:

$$\tau = \frac{L_{\text{eff},A \rightarrow B}}{h_{\text{RVE}}} \quad (13)$$

As previously stated, ASSB electrodes possess an almost zero porosity, which makes Eq. (12) not applicable to this specific battery technology. Within a conventional LIB electrode, ions flow through the liquid electrolyte that fills the pores. Hence, the porosity can be used to quantify the ionic conductivity. In the case of an ASSB electrode, the ions are transported through the solid electrolyte. Therefore, the amount of electrolyte directly influences the diffusion coefficient. Taking this into consideration, the porosity was substituted in Eq. (12) by the volume fraction of the electrolyte, an approach that has already been successfully taken in [11]. The electrolyte diffusion coefficient equals $2 \times 10^{-4} \text{ S cm}^{-1}$, as reported by Pesko et al. [39].

4 Results and Discussion

4.1 Calibration of the Electrical Resistivity of LFP and CB

Conductivity measurements of pure CB and LFP powders were performed as described in Sect. 2. These experiments were required in order to calibrate the electrical resistivity of both the LFP and CB particles. The corresponding experimental values and the porosities of the powders are presented in Tab. 3. Nevertheless, these experimental tests provide the electrical conductivity of the bulk material, but it is the electrical resistivity of the single particles that is required for the simulations, i.e., the ability of both the CB and LFP particles to resist the flow of electric current. These parameters are fundamental to the subsequent determination of the electrical conductivity of the electrode since the algorithm developed in this work takes into consideration the difference between the electrical properties of the two materials and stipulates the most conductive path accordingly.

Table 3. Experimental values of the electrical conductivity of the LFP and CB powders.

Parameter	LFP	CB
Porosity, ε [-]	0.60	0.74
Specific electrical conductivity, σ [S cm^{-1}]	0.023	4.390

Keeping this in mind, the experimental setup was reproduced in the DEM simulations for CB with a bulk porosity of 0.74, with the aim of calibrating the electrical resistivity of the particles. A total of 6000 particles with a particle size of 65 nm were simulated and the developed algorithm was used to calculate the total specific conductivity of the granular assembly. This value obtained numerically had to match the experimental one, and this was possible by fitting the electrical resistivity of

the particles. The final electrical resistivity of the CB particles was set at $0.07 \Omega \text{ mm}$. This procedure was also implemented for LFP, taking into consideration its particle size distribution and a bulk porosity of 0.60. The electrical resistivity of the particles required to fit the total specific conductivity of the granular sample was set at $6177.91 \Omega \text{ mm}$. These two values were used for the subsequent determination of the specific electrical conductivity of the ASSB electrodes by means of the implemented algorithm.

4.2 Validation of the Model: Variation in CB Content

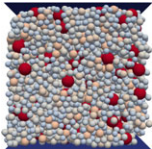
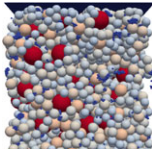
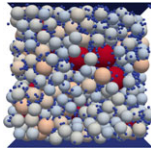
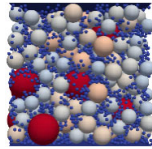
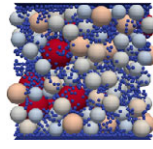
In order to validate the implemented method for assessing the electrical conductivity, five ASSB cathodes with different CB contents were produced and the electrical conductivity of each of them was measured following the procedure described in Sect. 2 (Tab. 4). Besides, the amount of electrolyte and the measured electrical conductivities are also presented in this table. Due to the variation in CB content, the amount of electrolyte (which in the simulation is taken as porosity, as already explained) slightly differs. As expected, an increment in CB is translated into an increase in electrical conductivity as a result of its high electrical conductivity in comparison to LFP. With respect to the DEM simulations, the electrodes represented the structural and mechanical properties of the real manufactured ones, as explained in Sect. 3. The total number of particles was aimed to be the same for all cases (6000 particles) in order to allow comparisons among the simulated structures. Thus, the simulation domain had to be adjusted accordingly. The length, width, and height of the simulation domain had the same values in each case. A snapshot of the electrodes generated numerically is also included in the table, as well as the size of the simulation domain. The colors of the particles correspond to their sizes, with the smallest particles (CB) being blue and the remaining ones representing the distribution of the LFP particles. As described in Sect. 3, the electrolyte was regarded as porosity in the simulations because of its insulating electrical properties. Thus, the electrolyte content presented in this table corresponds to the free space regarded in the DEM simulations.

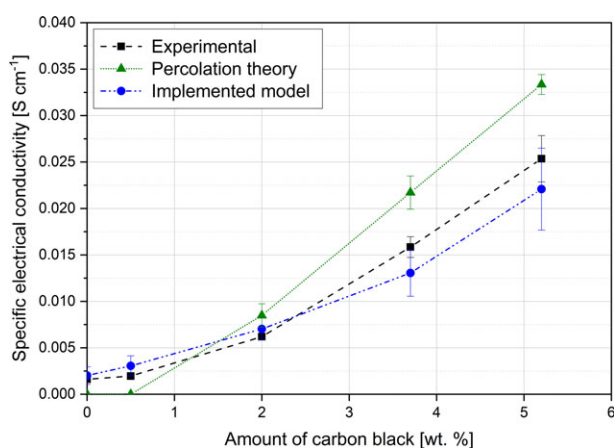
The simulations provided the necessary information about the electrode structures (distribution of CB and LFP particles, number of particle contacts, and contact area between two touching particles) that was required for the post-processing method. Furthermore, it was also possible to make use of these data in order to put into practice one of the most known theories for understanding the conductivity within composite materials: the percolation theory [21, 40, 41].

Fig. 4 shows the experimental specific electrical conductivity measured for the five cathodes, along with the one derived from the percolation theory and the specific electrical conductivity determined via the post-processing method proposed in this work.

As expected, incrementing the amount of CB leads to an increase in the electrical conductivity of the electrode. Considering the experimental data, at first, it can be seen that the specific conductivity increases moderately at low CB con-

Table 4. Properties of the manufactured and numerically generated cathodes.

	CB content [wt %]				
	0	0.5	2	3.7	5.2
Electrolyte content, V_{el} [%]	60.00	48.80	48.80	47.20	46.6
Experimental specific electrical conductivity, σ [$S\ cm^{-1}$]	1.61×10^{-3}	1.95×10^{-3}	6.20×10^{-3}	1.59×10^{-2}	2.54×10^{-2}
Simulation domain, $l = w = h$ [mm]	6.8	4.57	3.25	2.69	2.43
DEM structure					

**Figure 4.** Specific electrical conductivity: obtained experimentally, derived using the simulated data and the percolation theory, and determined via the simulated data and the implemented method.

tents and up to 2 wt %, from 1.61×10^{-3} to $6.20 \times 10^{-3} S\ cm^{-1}$. With a CB content greater than 2 %, the specific conductivity increases exponentially up to a value of $2.54 \times 10^{-2} S\ cm^{-1}$, for a CB content of 5.2 wt %. In order to explain this fact, the values obtained from the percolation theory can be of value. The percolation theory relies on the coordination number of the particles, i.e., the number of contacts a particle has with other neighboring particles [42,43], and establishes a relation between the structural parameters and the total electrical conductivity:

$$\sigma = \sigma_0((1 - \varepsilon)V_{CB}P_{CB})^\mu \quad (14)$$

where σ_0 represents the bulk electronic conductivity, ε accounts for the electrode porosity, and V_{CB} for the volume fraction of the electron conducting particles. The Bruggeman factor μ takes the effect of tortuous conduction paths into account and is typically set to 1.5 [42,44]. By considering the large differ-

ence in the intrinsic electrical conductivities of LFP and CB, it can be postulated that the electron conduction occurs mainly through CB particle contacts. Hence, the bulk electronic conductivity corresponded to that of CB. The percolation probability, P_{CB} , was defined from the average coordination number of CB particles, $Z_{CB,CB}$. In the past years, many percolation models have been proposed to study the conductivity within composite electrodes [41,42,45]. With the help of packing reconstruction methods and further imaging techniques, analytical expressions have been derived to relate this property with the microstructural parameters of the electrode. In this present work, the model of Costamagna et al. [46], supported by Chen et al. [42], is considered to determine the percolation probability, i.e., the probability that CB particles belong to a conductive path connecting both ends of the particulate configuration:

$$P_{CB} = \left[1 - \left(\frac{3.764 - Z_{CB,CB}}{2} \right)^{2.5} \right]^{0.4} \quad (15)$$

In general lines, the percolation theory establishes a critical concentration of conducting material that makes a transition from an insulating system to a conducting one. This critical concentration, above which conductive paths through the thickness of the particulate assembly can be found, is mostly known as the percolation threshold. Making reference to the electrical conductivity derived from the percolation theory (Fig. 4), the value of the electrode containing 0.5 wt % of CB equals zero, and so does the value for the electrode with no CB. This means that the percolation probability is below the percolation threshold. Thus, the composite electrodes may have isolated clusters of CB particles but not a complete percolated cluster that extends through the entire thickness of the analyzed simulation domain. Up to a critical concentration of CB (percolation threshold), effective conductive paths start to be built within the electrode and the conductivity is no longer zero. In fact, it is believed that the number of conductive paths increases significantly, which leads to a substantial increment in conductivity between the electrodes with 2 and 3.7 wt % of

CB. Following the percolation threshold, the conductivity keeps rising due to the increase in the number of conductive networks. Even though this theory can give a first estimation of the electrical conductivity, it overlooks some parameters that are also important for this property. In particular, the directionality of the contacts and the contact area are disregarded. Moreover, it only considers the electron conduction through CB particles, and this fact does not truly represent the reality since LFP is not an insulated material. In fact, LFP particles have a conductive coating through which the electrical transport is assured. On the contrary, the method proposed in this work incorporates these aspects. In fact, Fig. 4 clearly shows that the implemented model can capture the experimental trend more accurately. Even though the experimental tendency can be well represented by the model, there are some features that are not considered. In particular, all conductive paths are connected in parallel for the calculation of the specific electrical conductivity in the last part of the procedure, without regarding the effects of interconnected paths. This limitation of the model leads to an underestimation of the electrical conductivity, which becomes more relevant with an increase in particle contacts. As a consequence, the modeled values for the last two cathodes are slightly below the experimental values. This aspect will be taken into account in the near future in order to improve the precision of the model. However, it must be underlined that this method was aimed at giving a fast estimation of the specific electrical conductivity of ASSB cathodes. Besides, it can be used to assess the influence of structural and material parameters without the need to manufacture real electrodes. In fact, additional simulations were performed to further examine the progress of the electrical conductivity of the electrode influenced by the CB content. Tab. 5 presents the parameters regarding these computationally generated cathodes. The CB amount was varied from 8 to 20 wt %. The total number of particles was 6000 and the simulation domain was adjusted accordingly.

Fig. 5 shows the specific electrical conductivity of the simulated electrodes in dependence on the CB content. In general lines, an increase in this property can be observed along the increment in the CB concentration. Nevertheless, the increase is steeper at the beginning whereas it becomes gradual when the amount of CB is greater than approximately 8 wt %. In fact, a rise in the concentration of CB only from 2 to 3.7 wt % is translated into an increase in electrical conductivity of around 85.89 %, whereas incrementing the concentration of CB from 8

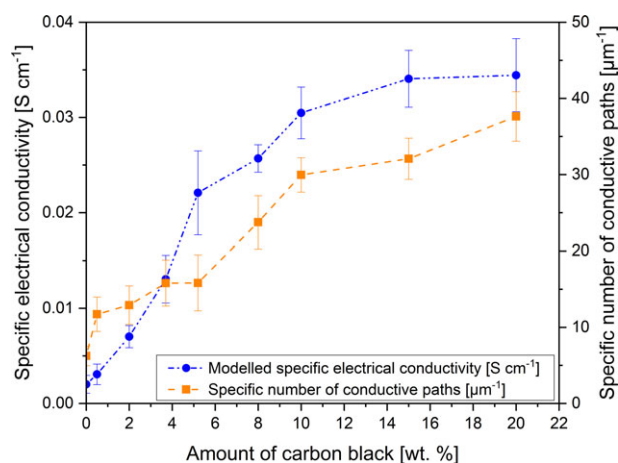


Figure 5. Modeled specific electrical conductivity and specific number of conductive paths as a function of the amount of CB for the simulated electrode structures.

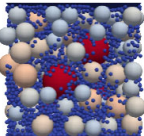
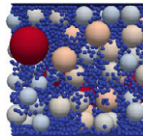
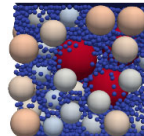
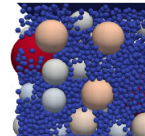
to 10 wt % implies an electrical conductivity increase of about 18.65 %. The formation of conductive paths throughout the electrode causes the initial steep increase in conductivity. Once the amount of CB is sufficient to complete the conductive network path within the structure, the increase in conductivity is much slower because it simply implies an increment in the number of networks. This event is known in the literature as the percolation transition and has been commonly observed in various fields of research [47–49].

In order to analyze the electrode structures in more detail, the specific number of conductive paths within each structure was also examined. A direct comparison of this value cannot be made since the simulation domains varied depending on the carbon content, with the purpose of guaranteeing the same number of particles. As a result, Fig. 5 shows the specific number of conductive paths calculated as the number of conductive paths multiplied by the height and divided by the surface area of the simulation domain:

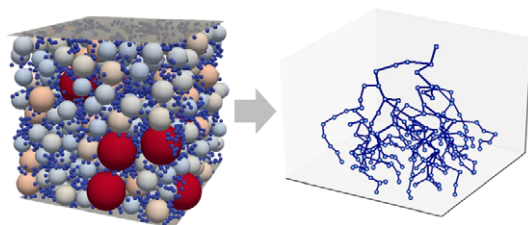
$$n_{\text{paths}}' = n_{\text{paths}} \frac{h_{\text{RVE}}}{w_{\text{RVE}} l_{\text{RVE}}} \quad (16)$$

It can be seen that, in general lines, the number of conductive paths increases along the amount of CB. Exemplarily, Fig. 6

Table 5. Additional cathodes generated numerically with a high CB content.

	CB content [wt %]			
	8	10	15	20
Electrolyte content, V_{el} [%]	45.00	43.00	45.00	46.00
Simulation domain, $l = w = h$ [mm]	2.19	2.05	1.83	1.69
DEM structure				

a) Simulated ASSB cathode with 3.7 %-wt. of CB



b) Simulated ASSB cathode with 15 %-wt. of CB

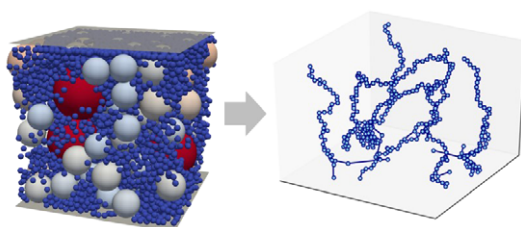


Figure 6. ASSB cathodes with (a) 3.7 wt % and (b) 15 wt % CB simulated via DEM, and their corresponding conductive paths obtained by the pathfinding algorithm.

depicts the connectivity within the electrodes with 3.7 and 15 wt % CB. On the left, and for both examples, the DEM-generated structure is shown with CB (smallest blue particles of 65 nm size) as well as LFP (polydisperse distribution and coarser in size) particles. On the right, the corresponding connectivity graph generated via the pathfinding algorithm is presented and used for the calculation of the electrical conductivity. This graph represents the position of the nodes (particles) with small points and the Euclidean distance between two nodes in contact with lines. It can be identified that the lines of the graph representing the connectivity of the electrode with 3.7 wt % CB are longer in comparison to the second one. This means that more LFP-LFP particle contacts were selected for the determination of the conductive paths, which explains the lower electrical conductivity. On the other hand, taking a look at the connectivity of the electrode with 15 wt % CB, it can be discerned that the nodes lie very close to each other, which means that they represent CB particles as they are smaller in size. Due to the fact that the concentration of CB particles is much higher within this second structure, the probability of selecting a CB instead of an LFP contact by the pathfinding algorithm is consequently greater. Therefore, the number of conductive paths is not the only relevant factor, but the quality of the particle contacts is also decisive for the calculation of the electrical conductivity.

Additionally, the tortuosity of all simulated structures was numerically calculated, and the effective ionic conductivity was derived using Eq. (12) and substituting the porosity for the electrolyte content as discussed in Sect. 3. The calculated values are presented in Fig. 7 along with the specific electrical conductivity of the simulated electrodes.

Given the electrical conductivity, it would not be advisable to increase the amount of CB above approximately 6–8 wt % because the increment in this property is not so steep above this point. On the other hand, at high concentrations of CB,

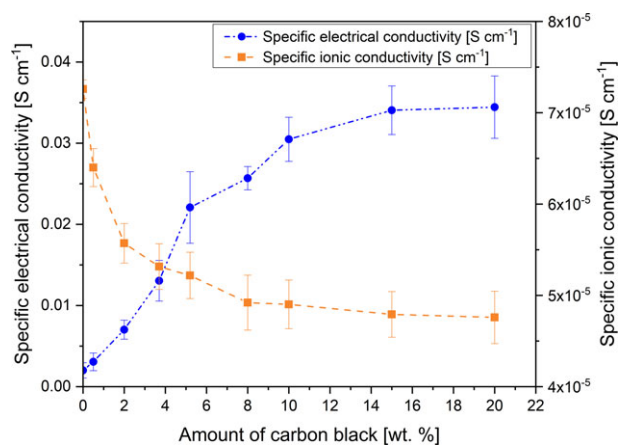


Figure 7. Modeled specific electrical conductivity and specific ionic conductivity as a function of the amount of CB for the simulated electrode structures

the physical presence of these particles leads to an increase in the tortuosity within the structure and, as a result, the ionic diffusion is hindered. This decrease in the ionic conductivity would imply a reduction in the electrochemical activity, which is not satisfactory for the performance of the ASSB. Consequently, in view of these outcomes and from the engineering point of view, an amount of CB of around 5 wt % is proposed to be a satisfactory compromise in order to reach suitable levels of ionic and electrical conduction.

5 Conclusion

In this work, a method is presented to determine the specific electrical conductivity within ASSB cathodes. The particulate structures are generated via the DEM by taking into consideration the particle properties of both materials and the electrolyte content within the cathode. Bearing in mind that the electron transport takes place through particle contacts, an analysis of the microstructure is performed subsequently in order to detect all contacts between and among the LFP and CB particles. Once the connectivity is assessed, the most conductive paths in *z*-direction are identified by means of a pathfinding algorithm improved for this purpose so that it accounts for the differences in the intrinsic electrical conductivities of the two materials. Subsequently, an equivalent circuit is defined regarding the particle contacts as resistances connected in series. A total equivalent resistance can be calculated by connecting all paths in parallel. The final step consists of determining the total specific electrical conductivity from the equivalent resistance by considering the size of the simulated domain as well. In order to calibrate the method, the electrical resistivities of the LFP and CB particles are determined experimentally and introduced in the code. Furthermore, real ASSB cathodes are manufactured with different CB contents and their electrical conductivities are measured. The results show that the implemented method can reproduce the experimental trend, even delivering a closer solution than the one derived from the percolation theory. This is possible by regarding all physical, mechanical,

and electrical characteristics, including the particle size, electrolyte content, and intrinsic material properties. The method is able to reproduce the increment and further consolidation of the electrical conductivity in dependence on the CB content, enabling its use for further investigations on electrode morphology and additional applications. Moreover, the tortuosity of the simulated electrodes is numerically calculated and the ionic conductivity is derived according to [11]. In fact, bearing in mind the gained outcomes, it is not advisable to increase the amount of CB above 5 wt %, due to the almost discernible increase in electrical conductivity and consequent hindering of ionic diffusion. Overall, this method offers the possibility to gain insights into the electrode connectivity in view of predicting the electrical conductivity and designing improved electrode structures. The next step of this research line is to further refine the method in order to be able to establish well-defined relationships between the structure and the electrode transport properties.

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Symbols used

A	[m ²]	contact area
D	[S m ⁻¹]	diffusion coefficient
$f(n)$	[-]	pathfinding function
$g(n)$	[-]	cost function
$h(n)$	[-]	heuristic function
h	[m]	height
l	[m]	length
n	[-]	node in the pathfinding algorithm
P	[-]	percolation probability
r	[m]	distance between two nodes/ particles
$\vec{r}$	[m]	position vector
V	[m ³]	volume
w	[m]	width
Z	[-]	coordination number
z	[m]	position of a node

Greek symbols

δ	[m]	overlap
ε	[-]	porosity
μ	[-]	Bruggeman factor
r	[S m ⁻¹]	resistivity
σ	[S m ⁻¹]	specific electrical conductivity
τ	[-]	tortuosity

Sub-/superscripts

cyl	cylinder connecting two contact points
current	current node in the pathfinding algorithm
eff	effective
el	electrolyte
goal	goal node
i	related to a particle
next	next node to be analyzed
par	sum of LFP and CB particles

Abbreviations

ASSB	all-solid-state battery
CB	carbon black
DEM	discrete-element method
LFP	lithium iron phosphate
LIB	lithium-ion battery
LiTFSi	lithium bis(trifluoromethylsulfonyl)imide
PEO	polyethylene oxide
RVE	representative volume element
SEM	scanning electron microscopy
SOFC	solid oxide fuel cell

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